The construction is a swimming pool tank type. Using 8.6 kg ^{235}U and 15 m^3 heavy water as moderator, it provides a maximum thermal neutron flux of 1.5×10^{19} m^{-2} s^{-1} at 60 MW_{th}; 10 fold higher fluxes can be achieved during shorter periods. At Oak Ridge, the High Flux Isotope Reactor (HFIR) has a flux of 3×10^{19} m^{-2} s^{-1}, the highest continuous thermal neutron flux so far reported. Some research reactors are built to test new reactor designs which may be candidates for future power reactors. The successful designs lead to prototype power reactors; some of these are described in Chapter 20 together with full-scale power reactors. Other designs never survive the first stage, although the design may be very interesting.

19.13. THERMAL POWER REACTORS

We illustrate the general principles of thermal reactors by a short description of the two most important power reactor types; the pressurized water reactor (PWR) and the boiling water reactor (BWR). They are further discussed in Chapter 20.

19.13.1. The pressurized water reactor

As the name implies a pressurized water reactor (PWR) is cooled by hot high pressure water, H_2O, which also acts as moderator. The main components of a pressurized light water moderated and cooled reactor (PWR) station have been described in §19.1 and Figure 19.2. The Ringhals station, whose first reactor started operation in 1975, is located in Sweden near the sea and seawater is used for cooling; power stations located near large rivers or at the sea shore usually do not need cooling towers.

The important features of the PWR core are shown in Figure 19.12, which, although taken from three different reactors, represents the typical Westinghouse design. A typical core contains ~40 000 fuel rods (a) in 193 assemblies, each with space for 208 fuel pins (b) and (c). The fuel is 88 t of UO_2 with an enrichment of 2.17% (inner region) to 2.67% (outer region) in the initial fuel loading. The reactor vessel (e) is made of stainless steel clad low alloy steel, 13 m high and 4.4 m in diameter, with a wall thickness of 22 cm; (c) shows the distribution of control rods entering from the top of the vessel. The 1060 control rods are made of a silver alloy containing 15 wt% In and 5 wt% Cd; both these elements have high thermal neutron capture cross sections. The main reactivity control is by boric acid, which is fed to the coolant through a special injection system. The boric acid circulates in the primary coolant loop, and acts as a neutron absorber. At the start of a fresh core its concentration is ~1500 ppm (~0.025 M), but it is successively reduced to zero at time of fuel replacement. The concentration is adjusted in a side loop, containing either an anion exchange or an evaporator system. The ion exchange system is so designed that H_3BO_3 is fixed to the ion exchanger (the water is deborated) at low temperature (~30°C) while it is eluted at higher temperature (~80°C).

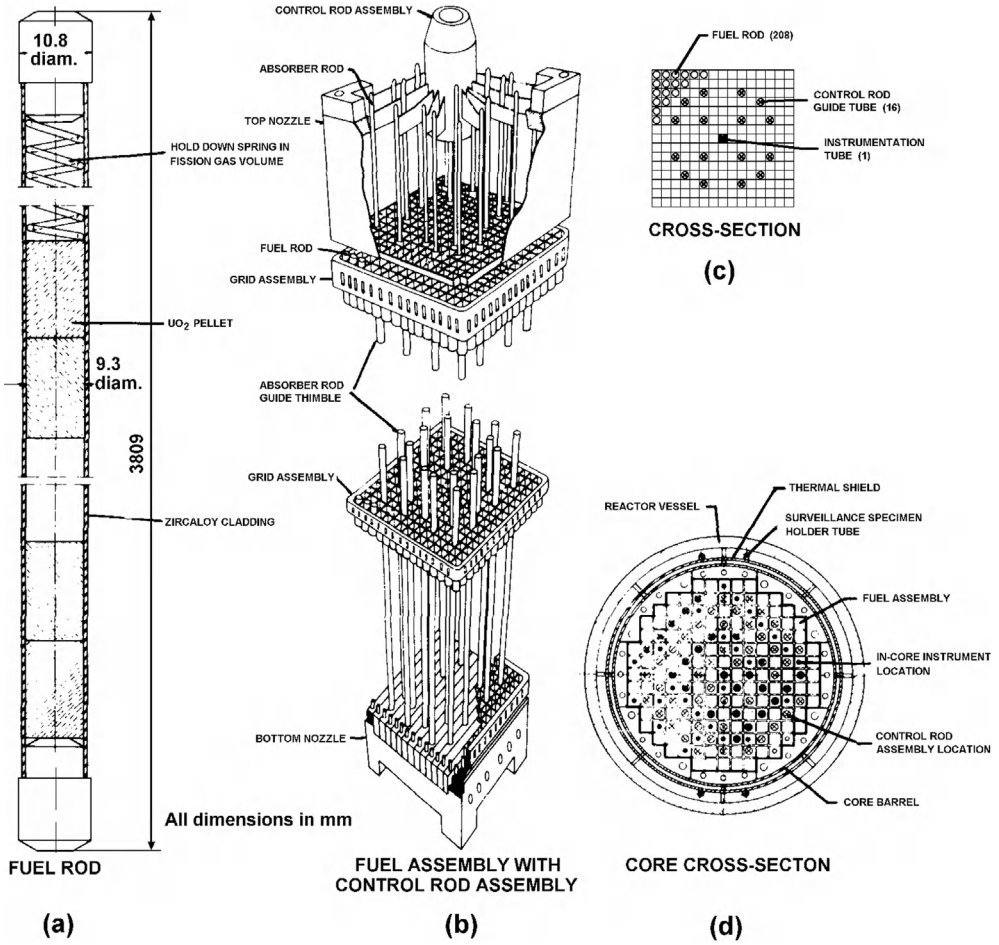


Figure 19.12 Typical PWR fuel pin (a), assembly (b, c), core (d), and vessel (e) designs.

Some of the physical parameters are: k_∞ 1.29 (cold) and 1.18 (operational temperature), temperature coefficient $-1 \times 10^{-3}\%\Delta k/k$ °C, average neutron flux 2.16×10^{17} (thermal), 3×10^{18} (fast) n m^{-2} s^{-1}. Data for heat production are given in Table 19.6.

The Westinghouse-type PWR (Figs. 19.2 and 19.12) reactor is presently the dominating nuclear power reactor in the world.

19.13.2. The boiling water reactor

In a boiling water reactor (BWR) the coolant is brought to boiling while passing along the fuel pins. Thus the need for a separate steam generator and secondary loop is removed. This gives a robust design with less complexity and marginally higher thermal efficiency than a PWR at the expense of some features. To minimize the wear of the

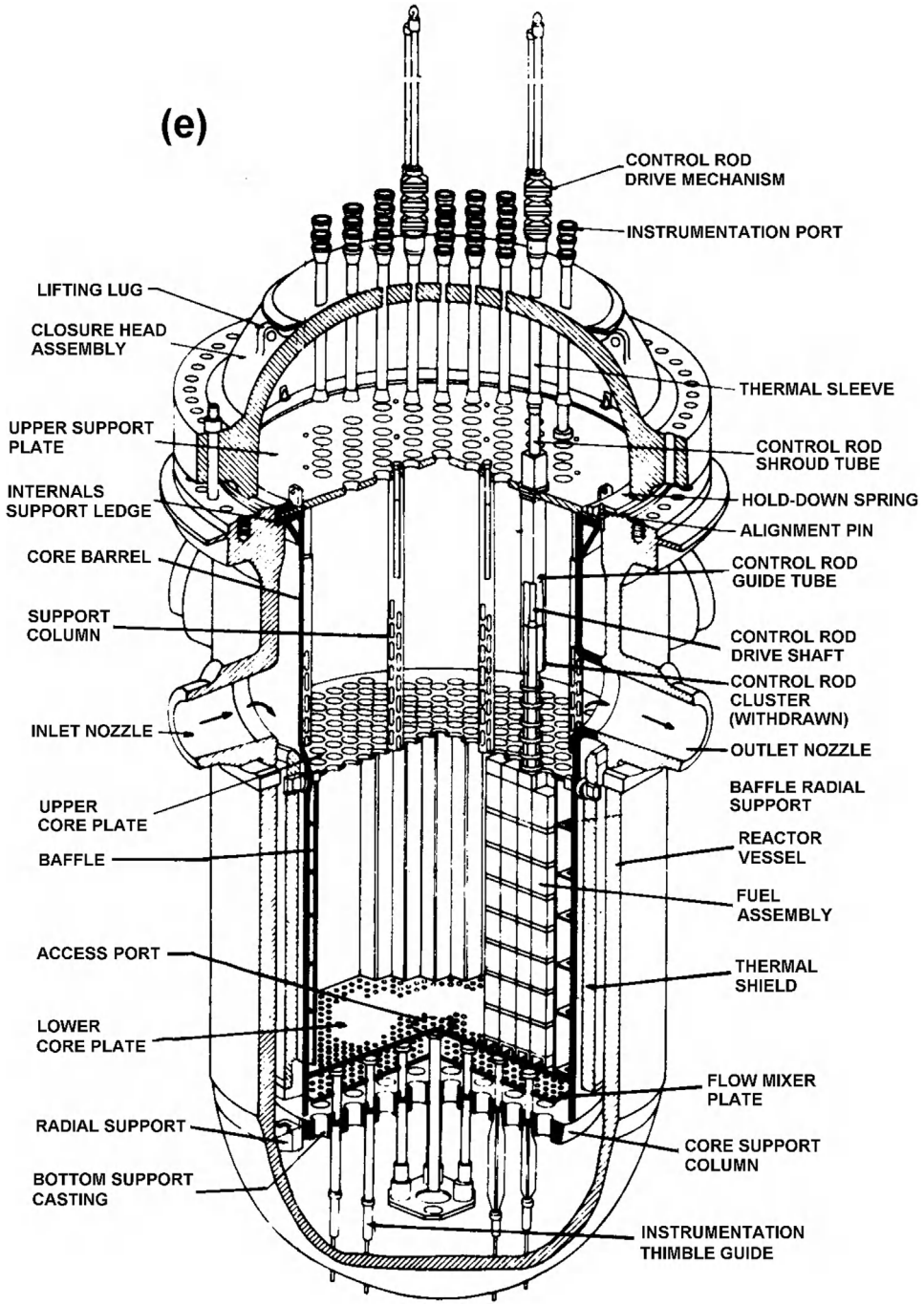


Figure 19.12 (*continued*).

Table 19.6 The international nuclear event scale (INES) used for the reporting of nuclear events

Type	Definition	Value	Examples	
None	No safety significance	0		
Incident	Anomaly	1		
	Incident	2		
	Serious incident	3	Vandellos	1989
Accident	Accident mainly in installation	4	Saint Laurent	1980
	Accident with off-site risks	5	TMI	1979
	Serious accident	6		
	Major accident	7	Chernobyl	1986

turbine it must be fed with steam that is as dry as possible. For this purpose steam separators and steam dryers are needed. The water not converted to steam is recirculated back into the reactor core by external or internal recirculation pumps.

The single coolant − steam loop permits shortlived activity from neutron activation of oxygen in the coolant to reach the turbine. The main reaction with thermal neutrons is

$$^{18}\mathrm{O}\,(\mathrm{n},\gamma)\,^{19}\mathrm{O} \xrightarrow[27.1\ \mathrm{s}]{\beta,\gamma} {}^{19}\mathrm{F} \tag{19.40}$$

and with high energy neutrons

$$^{16}\mathrm{O}\,(\mathrm{n},\mathrm{p})\,^{16}\mathrm{N} \xrightarrow[7.13\ \mathrm{s}]{\beta,\gamma} {}^{16}\mathrm{O} \tag{19.41}$$

Small amounts of fission products leaking from faulty fuel pins or generated from uranium (leached from damaged fuel or initially present as impurity) deposited on the surfaces in the core (so called *tramp uranium*) and small amounts of activation products (most troublesome is ^{60}Co) are carried by water droplets from the reactor vessel through the piping to turbine and condenser. The presence of γ-emitters in the steam makes it necessary to shield pipes and turbines and limit access to the turbine hall during operation.

The fact that the boiling water never leaves the reactor system makes it necessary to have a special reactor water cleaning system attached. Usually this system contains mixed-bed ion exchangers operating on partially cooled (~90°C) reactor water. To further reduce the accumulation rate of corrosion products in the reactor, the water from the turbine condenser is also cleaned by ion exchange filters before being returned as feed water to the reactor.

The reactor core and vessel of a typical boiling light water reactor station, a BWR, is shown in Figure 19.13 and supplemental data are given in Table 19.4. The fuel core and elements are rather similar to those of the PWR. The main differences are that boiling

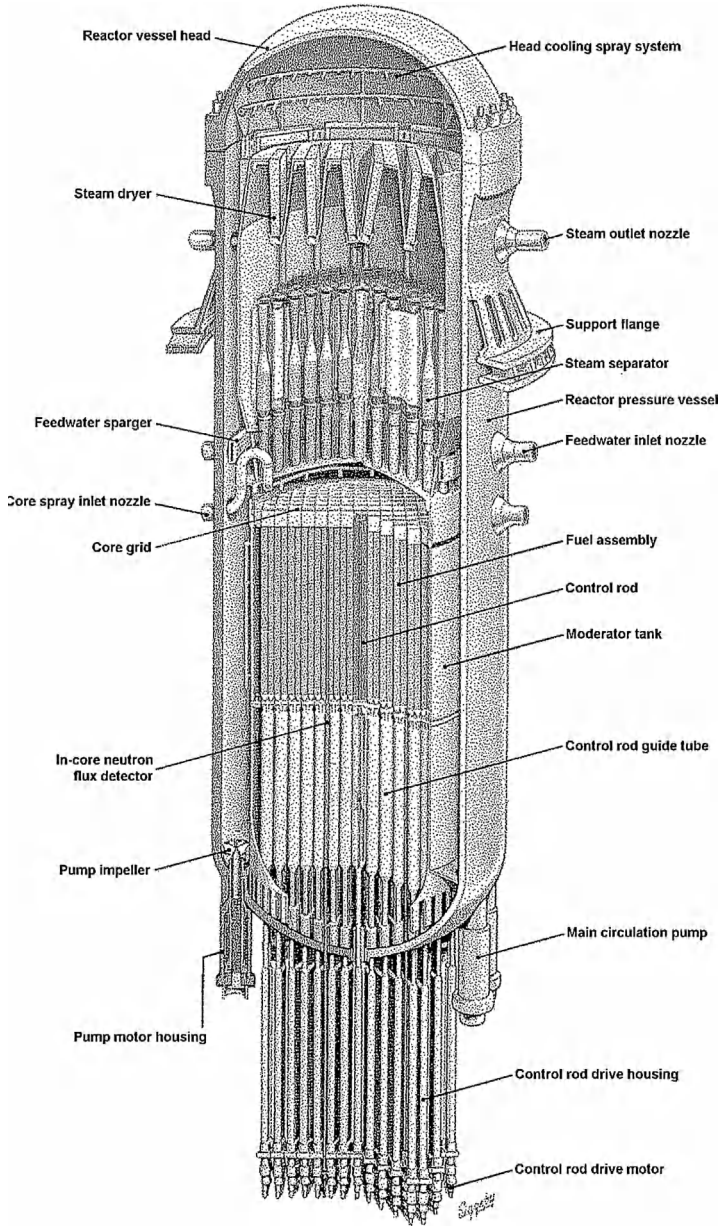


Figure 19.13 Design of a boiling water reactor core system, ABB-Atom, Sweden.

occurs in the reactor vessel, so no external steam generator is required, and that the fuel elements are enclosed in fuel boxes to improve the flow pattern. Although the cores are about the same size for PWRs and BWRs, the reactor vessel is much taller for the BWR; typical values are 22 m in height and 6 m in diameter. Since the pressure in the vessel is lower for BWRs than for PWRs, the vessel wall can be thinner (a typical value is 16 cm). BWRs usually contain more uranium, but with lower enrichment, than PWRs. While only about one-fifth of the core is replaced each year in a BWR, about one-third to one-fourth of the core is replaced annually for a PWR; in both cases this corresponds to about 30 t of used fuel replaced every year.

The BWR is controlled by blades entering from the bottom of the reactor vessel. The blades are usually of cruciform cross section and fit in between the fuel boxes surrounding the fuel elements. The fuel elements often contain burnable poison (e.g. Gd_2O_3). Because of the boiling in the tank boric acid cannot be used for reactivity control; however, it is provided as an extra emergency shutdown feature. The control blades contain boron carbide rods or powder placed in channels in the stainless steel blade-wings. In addition to negative temperature coefficients both for fuel and moderator, BWRs have a negative *void coefficient*. This means that bubble formation (*void*) along the fuel pins reduces the reactivity. As mentioned in §19.6 it is therefore common practice to control the reactor power by the main coolant circulation pumps; increased circulation initially reduces the voids (and fuel temperature) thereby increasing power production until the previous void fraction has been almost restored at a slightly higher average fuel temperature. However, a very large negative void coefficient should also be avoided as it can cause an undesirable amplification of pressure fluctuations in the reactor vessel.

19.14. POWER STATION EFFICIENCY

Not all of the heat produced in a nuclear reactor can be used for work, i.e. for turning the turbine blades connected to the rotor of an electric generator. According to the second law of thermodynamics, the maximum *thermal efficiency* (η) is

$$\eta = (Q_{in} - Q_{out})/Q_{in} \tag{19.42}$$

where Q_{in} is the heat input into a machine and Q_{out} is the heat discharged from that machine. Most reactors operate according to the *Rankine cycle* in which a liquid medium is heated and vaporized at constant pressure, and work is carried out through adiabatic expansion. While the steam entering the turbine should be as dry as possible to reduce turbine blade wear, the steam on the backside is wet due to some condensation. This effect is normally reduced by dividing the turbine into a high pressure and a low pressure part with intermediate reheating by primary steam. Because steam is not an ideal gas (an accurate calculation of Q from heats of vaporization and the ideal gas law is impossible) empirically based steam tables must be used.